

6DOF Arm — Wiring Bench Sheet

ESP32 + PCA9685 + 6× MG996R · print & keep at the bench · full guide: [hardware/wiring.md](#)

SAFETY RULES — read before plugging anything in

- 1. NEVER put 12 V on the PCA9685 V+ terminal or any servo.** MG996Rs are rated 4.8–7.2 V. The 12 V supply feeds **ONLY** the converter input.
- 2. USB power = single-servo bench tests only.** Full-arm motion requires the 6 V/10 A rail.
- 3. Verify converter output ≈ 6.0 V with a meter BEFORE first connection.** Re-check polarity at V+ — reverse polarity kills the PCA9685 instantly.
- 4. Converter → PCA9685 wires short and thick** (14–16 AWG). Stalls can pull 15 A+.
- 5. Power-up order:** ESP32 USB first (servos stay idle), then the 12 V supply. Reverse to power down.
- 6. Pinch points:** on link loss the arm **HOLDS** (won't drop, won't yield). Kill 12 V before handling.

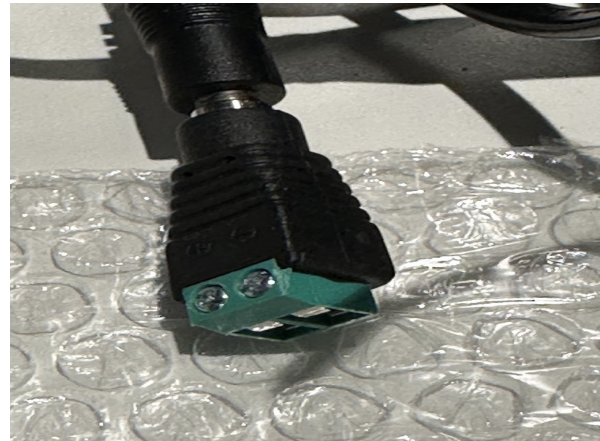
Servo power path

12 V/10 A supply → buck converter (12–40 V in, 6 V/10 A out) → PCA9685 V+ terminal

Converter wire colors — straight from the unit's label

Wire	Role	Connects to
Red	Input +	12 V supply positive
Black (paired w/ red)	Input –	12 V supply negative
Yellow	Output + (6 V)	PCA9685 V+ screw terminal
Black (paired w/ yellow)	Output –	PCA9685 GND screw terminal

The two blacks are per-bundle negatives — use each with its own side. Red side drinks 12 V; yellow side feeds the servos. The PCA9685's onboard 1000 μ F cap covers the rail; no extra cap needed.



Left: the converter's own label (IN red/black, OUT yellow/black). Right: barrel-jack → screw-terminal adapter for the 12 V input — mind the +/– marks.

Logic side: ESP32 ↔ PCA9685 (I2C)

PCA9685 header order, starting from the end AWAY from the silver capacitor: **GND, OE, SCL, SDA, VCC, V+** (V+ is always the pin beside the cap). Ribbon colors below are this build's harness as photographed. I2C address 0x40, jumpers open; board pull-ups are to VCC — which is why VCC must be 3.3 V.

Ribbon color	PCA9685 pin	ESP32 pin	Notes
brown/black (end pin)	GND	GND	Required — the common-ground link; I2C is dead/flaky without it
red	OE	leave unplugged	Already pulled low (enabled); future kill-switch option
orange	SCL	D22 (GPIO22)	Arduino Wire default clock
yellow	SDA	D21 (GPIO21)	Arduino Wire default data
green	VCC	3V3	3.3 V only — never the 5 V/VIN pin
blue (beside cap)	V+	REMOVE — never to ESP32	6 V servo rail! Same net as the terminal block; kills the ESP32

ESP32 pin locations (30-pin devkit): USB-C toward you, antenna away — right-hand row from the USB end reads 3V3, GND, D15, D2, D4, RX2, TX2, D5, D18, D19, **D21**, RX0, TX0, **D22**, D23. Green + brown at the near corner; yellow on D21 (11th); orange on D22 (14th).

Servo plugs and channels

Column pin order from board edge inward: **GND (bottom), V+ (middle), PWM (top)**. MG996R leads: **brown = GND, red = V+, orange = signal** — brown faces the board edge.

Channel	Joint	Position on arm
0	joint1	Base rotation (yaw)
1	joint2	Shoulder pitch
2	joint3	Elbow pitch
3	joint4	Wrist pitch
4	joint5	Wrist roll
5	joint6	Gripper
6–15	—	Spare

Wrong channel? Move the plug, don't edit firmware — keeps this table true.

Common ground — required, not optional

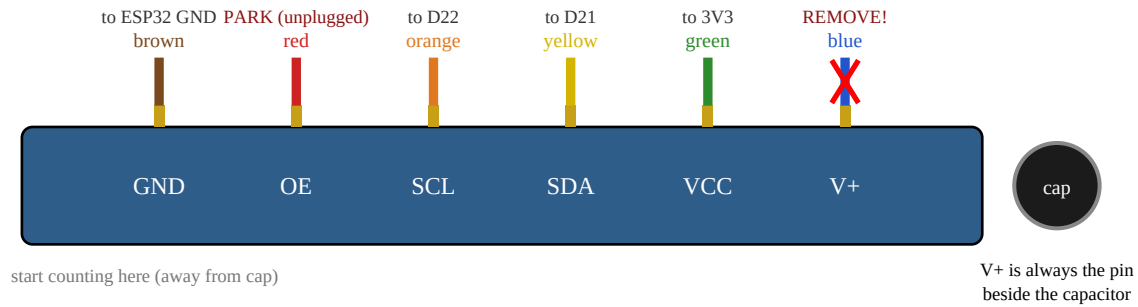
PSU(–) → converter IN(–) → converter OUT(–) → PCA9685 GND → ESP32 GND (via the I2C ground jumper). Beep-test all four points.

Bench bring-up checklist

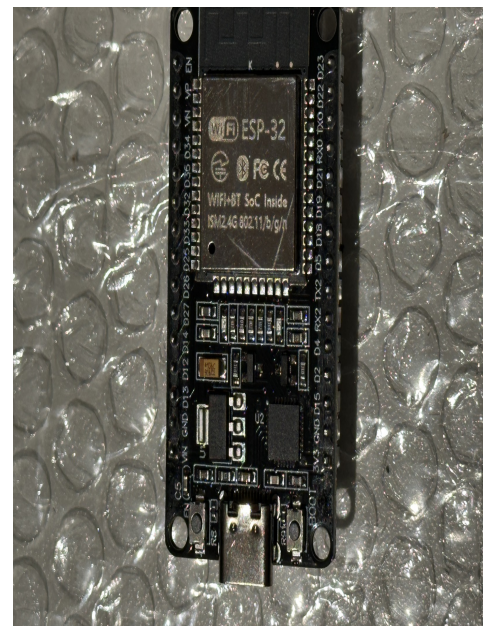
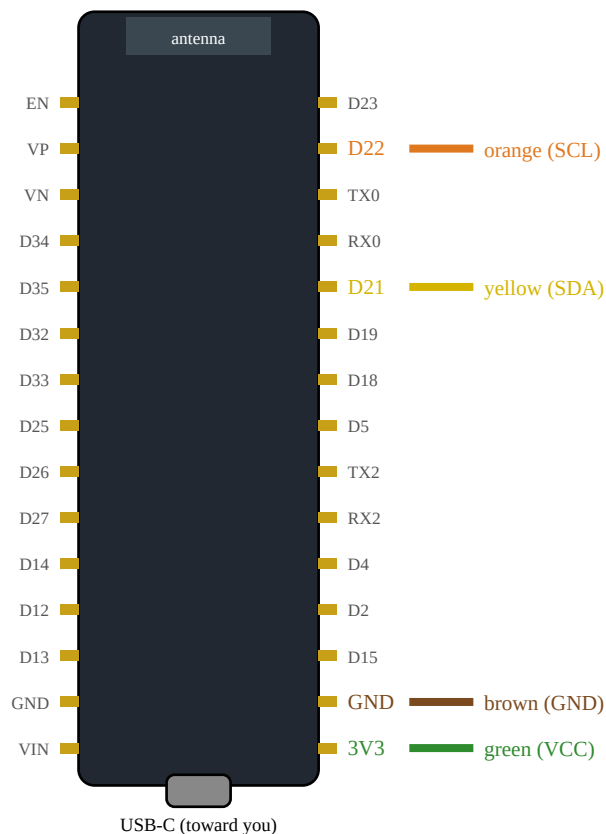
- [] All grounds common (continuity-beep PSU–, converter–, PCA9685 GND, ESP32 GND)
- [] Converter output ≈ 6.0 V, verified with meter, no load
- [] I2C jumpers by color: green→3V3 · yellow→D21 · orange→D22 · brown→GND
- [] Red (OE) jumper parked; blue (V+) jumper REMOVED from ribbon
- [] ESP32 on USB: serial prints “PCA9685 initialized, all channels off”
- [] One servo on channel 0 — single-servo sweep (USB/bench power OK)
- [] Remaining servos on channels 1–5 per table
- [] Full-arm motion ONLY on converter power

Pinout diagrams

PCA9685 I2C header — identify direction by the capacitor:



ESP32 (30-pin devkit) — the four connections, highlighted:



Left: schematic top view (USB-C toward you). Right: the actual board — match the silk labels; D-numbers are GPIO numbers.